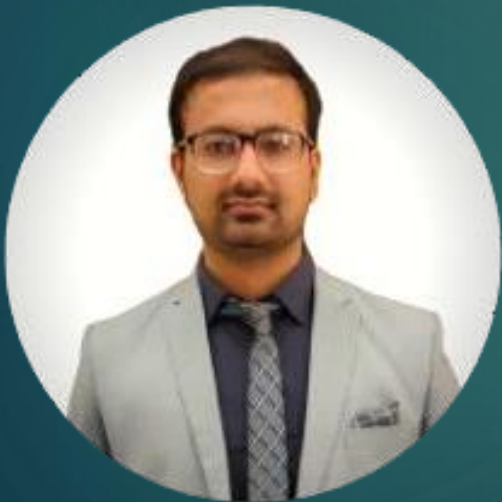


MOBILE AND APPLICATION DEVELOPMENT

INTRODUCTION



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android 

What is Android?



- Mobile operating system based on Linux kernel
- User Interface for touch screens
- Used on over 80% of all smartphones
- Powers devices such as watches, TVs, and cars
- Over 2 Million Android apps in Google Play store
- Highly customizable for devices / by vendors
- Open source

Developed by the **Open Handset Alliance (OHA)** led by **Google** including **80 + Companies**

WHY



statcounter
GlobalStats

Android

72.92%

iOS

26.53%

Samsung

0.22%

Unknown

0.13%

Mobile Operating System Market Share Worldwide

Prerequisites

1. Java
2. Kotlin

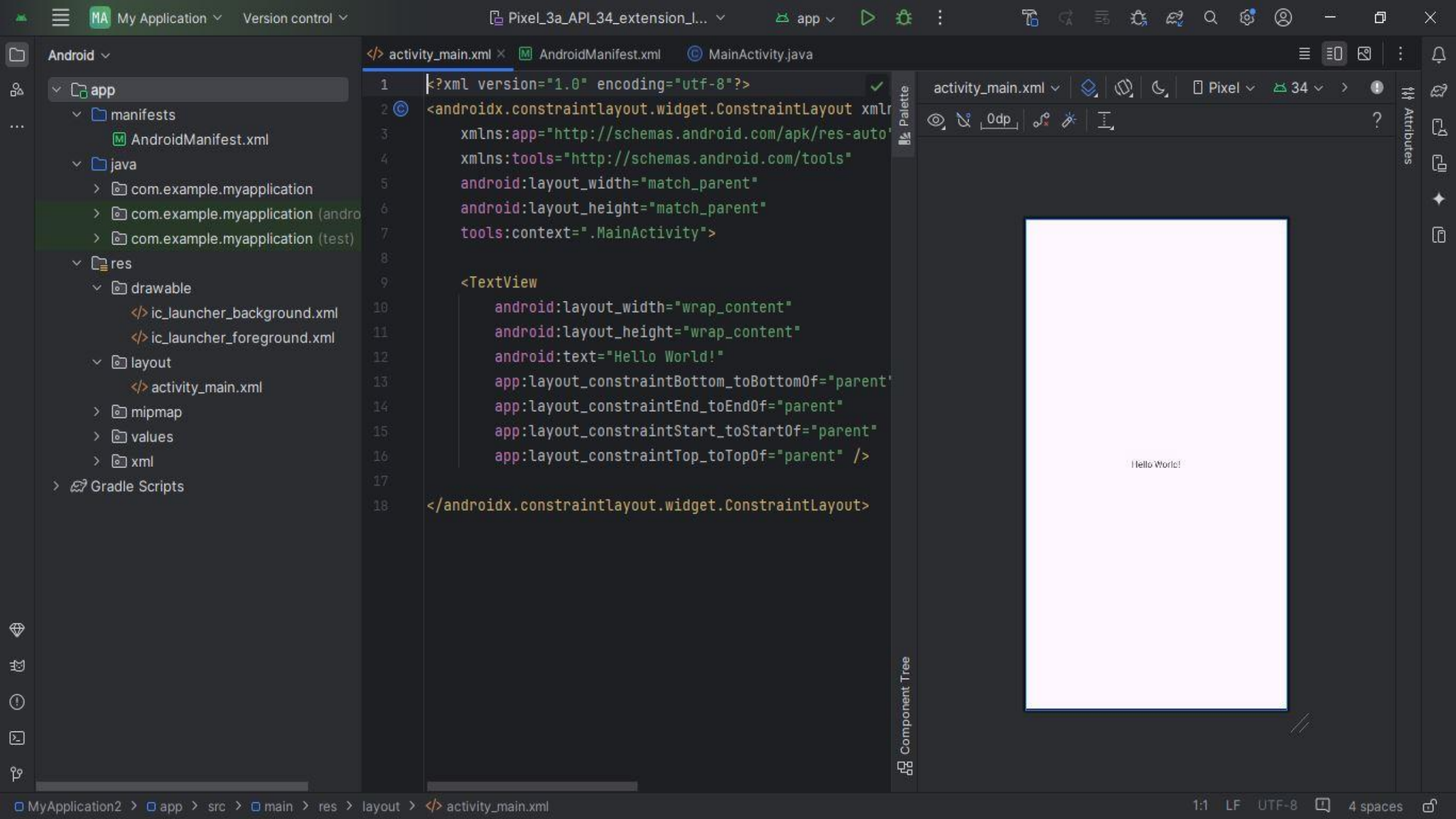
- Java Programming Language
- Object-oriented programming
- XML - properties / attributes
- Using an IDE for development and debugging



Integrated Development Environment

Main Features of IDE

- Editing Source Code
- Syntax Highlighting
- Autocomplete (Suggestions)
- Building Executables
- Debugging (Manual Testing)



Android Versions



Name	Version number(s)	Initial stable release date	Supported (security fixes)	API level
No official codename	1.0	September 23, 2008	No	1
	1.1	February 9, 2009	No	2
Cupcake	1.5	April 27, 2009	No	3
Donut	1.6	September 15, 2009	No	4
Eclair	2.0 – 2.1	October 26, 2009	No	5 – 7
Froyo	2.2 – 2.2.3	May 20, 2010	No	8
Gingerbread	2.3 – 2.3.7	December 6, 2010	No	9 – 10
Honeycomb	3.0 – 3.2.6	February 22, 2011	No	11 – 13
Ice Cream Sandwich	4.0 – 4.0.4	October 18, 2011	No	14 – 15
Jelly Bean	4.1 – 4.3.1	July 9, 2012	No	16 – 18
KitKat	4.4 – 4.4.4	October 31, 2013	No	19 – 20
Lollipop	5.0 – 5.1.1	November 12, 2014	No	21 – 22
Marshmallow	6.0 – 6.0.1	October 5, 2015	No	23
Nougat	7.0 – 7.1.2	August 22, 2016	No	24 – 25
Oreo	8.0 – 8.1	August 21, 2017	Yes	26 – 27
Pie	9	August 6, 2018	Yes	28
Android 10	10	September 3, 2019	Yes	29
Android 11	11	September 8, 2020	Yes	30

Android user interaction

- Touch gestures: swiping, tapping, pinching
- Virtual keyboard for characters, numbers, and emoji
- Support for Bluetooth, USB controllers and peripherals

Google Play store

Publish apps through [Google Play](#) store:

- Official app store for Android
- Digital distribution service operated by Google



Challenges of Android development

- Multiple screen sizes and resolutions
- Performance: make your apps responsive and smooth
- Security: keep source code and user data safe
- Compatibility: run well on older platform versions
- Marketing: understand the market and your users
(Hint: It doesn't have to be expensive, but it can be.)